Lecture 2

Noether currents

The first-variation formula separates the field equation from a divergence. A continuous symmetry gives another expression for the same variation. Comparing the expressions produces a local conservation law. Integrating that law requires a separate condition on the spatial boundary.

Objectives. *We prove Noether's identity, including transformations that move spacetime points. We compute the current for rotations of a complex field. We state and prove sufficient conditions for its spatial integral to be constant.*

By the end of this lecture you will be able to:

- Compute the characteristic of an internal symmetry or a spacetime translation.
- Derive a Noether current and its identity before imposing the field equation.
- Compute the complex Klein–Gordon current with the course normalisation.
- Prove charge conservation on the torus and under stated decay on $\mathbb{R}^d$.

2.1 Currents and infinitesimal changes

We retain the coordinates, C^2 fields and C^2 Lagrangian densities of Lecture 1. Let $I \subset \mathbb{R}$ be an open time interval. Space is either $X = \mathbb{T}^d$ or $X = \mathbb{R}^d$.

Definition 2.1 (Current and local conservation law)**.** A *current* on $I \times X$ is a C^1 map

$$j\colon I \times X \longrightarrow \mathbb{R}^{d+1}, \qquad (t,x) \longmapsto \big(j^0(t,x), \dots, j^d(t,x)\big).$$

The current satisfies a *local conservation law* if

$$\partial_\mu j^\mu = \partial_t j^0 + \partial_i j^i = 0.$$

The component j^0 is the density. The components j^i describe spatial flux. Neither the sign of j^0 nor the convergence of its spatial integral is part of this definition.

To compare a field with its transform, we must specify whether the spacetime point moves. Consider a C^2 family of local point transformations, with parameter ε near zero, given by

$$x' = f_\varepsilon(x), \qquad u' = g_\varepsilon(x,u), \qquad f_0(x) = x, \quad g_0(x,u) = u.$$

Each f_ε is a local diffeomorphism. On the image of an open set where f_ε is one-to-one, the transformed field is defined by

$$\phi_\varepsilon\big(f_\varepsilon(x)\big) = g_\varepsilon\big(x, \phi(x)\big).$$

We take the family to be C^2 jointly in its arguments. Its infinitesimal coefficients are $\xi^\mu(x) = \partial_\varepsilon f_\varepsilon^\mu(x)|_0$ and $\zeta^a(x,u) = \partial_\varepsilon g_\varepsilon^a(x,u)|_0$.

Definition 2.2 (Characteristic)**.** The *characteristic* of this family, evaluated on a field ϕ, is its change at a fixed spacetime point:

$$Q^a[\phi](x) := \partial_\varepsilon \phi^a_\varepsilon(x)\big|_0 = \zeta^a\big(x, \phi(x)\big) - \xi^\nu(x)\partial_\nu \phi^a(x). \tag{2.1}$$

A transformation with $f_\varepsilon = \mathrm{id}$ is an *internal transformation*.

The chain rule applied to the defining equation for ϕ_ε proves (2.1). For an internal transformation, the characteristic is simply $\zeta^a(x, \phi(x))$. For the translation $f_\varepsilon(x) = x + \varepsilon e_\nu$, with $g_\varepsilon(x, u) = u$, it is $Q^a = -\partial_\nu \phi^a$. Here e_ν is the coordinate vector in the x^ν direction.

Although ϕ is C^2, its characteristic need only be C^1. Lecture 1 allows C^1 variation directions. The local first-variation formula therefore applies with $h^a = Q^a$.

2.2 Symmetry up to a divergence

Write $\delta_Q L[\phi]$ for the derivative of the density at a fixed point along ϕ_ε. The chain rule gives

$$\delta_Q L[\phi] = \frac{\partial L}{\partial u^a}[\phi] Q^a + p^\mu_a \partial_\mu Q^a, \qquad p^\mu_a = \frac{\partial L}{\partial v^a_\mu}[\phi].$$

Every derivative ∂_μ of an evaluated expression below is a derivative of a composite function on spacetime. It differentiates through ϕ and its derivatives. A divergence is always understood in this sense.

Definition 2.3 (Variational and divergence symmetry)**.** The family is an infinitesimal *variational symmetry* if, for every C^2 field, there are specified local expressions $B^\mu[\phi]$ such that their evaluations are C^1 and

$$\delta_Q L[\phi] = \partial_\mu B^\mu[\phi]. \tag{2.2}$$

Allowing a nonzero divergence on the right is called a *divergence symmetry*.

This definition concerns the infinitesimal change. It requires an identity for arbitrary fields, without imposing the Euler–Lagrange equations. The expressions B^μ must be given by the transformation and density, rather than chosen separately by integrating for each field.

For a point transformation, one often knows the change of the density including the coordinate volume. Define its coefficient $\mathcal{A}[\phi]$ by

$$\mathcal{A}[\phi] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_0 \left\{ L[\phi_\varepsilon]\big(f_\varepsilon(x)\big) \det Df_\varepsilon(x) \right\}.$$

The derivative of the determinant at the identity is $\partial_\mu \xi^\mu$. Differentiating the other factor at its moving argument gives $\delta_Q L + \xi^\mu \partial_\mu L[\phi]$. The product rule consequently gives

$$\mathcal{A}[\phi] = \delta_Q L[\phi] + \partial_\mu(\xi^\mu L[\phi]). \tag{2.3}$$

Thus a point transformation with $\mathcal{A} = \partial_\mu b^\mu$, where the evaluated expressions b^μ are C^1, gives (2.2) with $B^\mu = b^\mu - \xi^\mu L[\phi]$. In particular, exact invariance of the density including its volume means $\mathcal{A} = 0$, and then $B^\mu = -\xi^\mu L[\phi]$. The point divergence b^μ and the fixed-point divergence B^μ differ. Exact invariance at moving points therefore need not give $\delta_Q L = 0$ at fixed points.

Theorem 2.4 (Noether's identity · *proved*)**.** *Let L and ϕ be C^2. Suppose a C^1 characteristic $Q^a[\phi]$ and C^1 expressions $B^\mu[\phi]$ satisfy (2.2) for every field. The* Noether current *is*

$$j^\mu = p^\mu_a Q^a - B^\mu. \tag{2.4}$$

It satisfies the identity

$$\partial_\mu j^\mu = -E_a[\phi]Q^a, \qquad E_a[\phi] = \frac{\partial L}{\partial u^a}[\phi] - \partial_\mu p^\mu_a. \tag{2.5}$$

On a solution, $E_a[\phi] = 0$, the current satisfies a local conservation law.

Proof. We use the product rule in the local first-variation formula:

$$\partial_\mu B^\mu = \delta_Q L = E_a Q^a + \partial_\mu (p^\mu_a Q^a).$$

Subtracting the divergence of B^μ proves (2.5). The products are C^1, so every derivative used here exists classically. On a solution the right-hand side is zero. ∎

Our Euler–Lagrange expression has the sign fixed in Lecture 1. The current sign follows from that choice and (2.4). Olver's statement writes $\operatorname{Div} P = Q^a E_a$, so his current is $P^\mu = -j^\mu$.[1]

Example 2.5 (A nonzero divergence term)**.** For one real field take the density

$$L = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi + \phi\phi_t.$$

Under the internal shift $\phi_\varepsilon = \phi + \varepsilon$, the characteristic is $Q^1 = 1$. Its density change is $\delta_Q L = \phi_t$. Thus $B^0 = \phi$ and $B^i = 0$. The momentum has components $p^0 = \phi_t + \phi$ and $p^i = -\partial_i\phi$. The current is therefore $j^\mu = \partial^\mu\phi$. The divergence term removes the extra ϕ from its time component.

2.3 Rotations of a complex Klein–Gordon field

Let $\Phi = \phi_1 + \mathrm{i}\phi_2$, with real C^2 components. Take $m \geq 0$ and the density

$$L = \frac{1}{2}\partial_\mu\overline{\Phi}\,\partial^\mu\Phi - \frac{1}{2}m^2|\Phi|^2 = \frac{1}{2}\sum_{a=1}^{2}\left(\partial_\mu\phi_a\,\partial^\mu\phi_a - m^2\phi_a^2\right). \tag{2.6}$$

There is no factor $1/\sqrt{2}$ in the definition of Φ. The displayed factor $1/2$ gives each real component the normalisation of Lecture 1. We differentiate with respect to those two real components.

The group $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$ acts by internal rotations $\Phi_\varepsilon = e^{\mathrm{i}\varepsilon}\Phi$. Its real-component form gives

$$\begin{pmatrix}\phi_{1,\varepsilon}\\ \phi_{2,\varepsilon}\end{pmatrix} = \begin{pmatrix}\cos\varepsilon & -\sin\varepsilon\\ \sin\varepsilon & \cos\varepsilon\end{pmatrix}\begin{pmatrix}\phi_1\\ \phi_2\end{pmatrix}, \qquad (Q^1, Q^2) = (-\phi_2, \phi_1).$$

Since the parameter is constant, the derivatives rotate by the same matrix. That matrix preserves the Euclidean scalar product in the two components. Both terms in (2.6) are invariant. Hence $\delta_Q L = 0$ and we take $B^\mu = 0$.

[1] Olver, *Applications of Lie Groups to Differential Equations*, §4.4, Theorem 4.29 and equations (4.38)–(4.41), pp. 272–274. The course notation bible records this translation.

The momenta are $p_a^\mu = \partial^\mu \phi_a$. Substitution in (2.4) gives the current

$$j^\mu = \phi_1 \partial^\mu \phi_2 - \phi_2 \partial^\mu \phi_1 = \operatorname{Im}(\overline{\Phi}\, \partial^\mu \Phi). \tag{2.7}$$

Its components are $j^0 = \operatorname{Im}(\overline{\Phi}\Phi_t)$ and $j^i = -\operatorname{Im}(\overline{\Phi}\partial_i \Phi)$. The minus sign in the spatial components comes from the Minkowski metric.

The field equations are $(\Box + m^2)\phi_a = 0$ for $a = 1, 2$. We can also verify Noether's identity directly. The two products of first derivatives cancel, so

$$\partial_\mu j^\mu = \phi_1 \Box \phi_2 - \phi_2 \Box \phi_1 = \phi_1(\Box + m^2)\phi_2 - \phi_2(\Box + m^2)\phi_1 = -E_a Q^a.$$

On solutions each term containing $(\Box + m^2)\phi_a$ is zero, so the divergence vanishes.

For a specific solution on $\mathbb{R} \times \mathbb{T}$, take $m = 3$ and $\Phi(t, x) = Ae^{-\mathrm{i}\sqrt{13}t + 2\mathrm{i}x}$, where $A \in \mathbb{C}$ is constant. Its second derivatives give $(\Box + 9)\Phi = (-13 + 4 + 9)\Phi = 0$. Its current has the constant values

$$j^0 = -\sqrt{13}|A|^2, \qquad j^1 = -2|A|^2.$$

We call time dependence $e^{-\mathrm{i}\nu t}$ with $\nu > 0$ *positive frequency*. Thus this positive-frequency solution has negative density for the chosen rotation $\Phi \mapsto e^{\mathrm{i}\varepsilon}\Phi$. Reversing the rotation reverses the current.

2.4 From a local law to a charge

Definition 2.6 (Charge)**.** For a current whose time component is integrable on each slice, its *charge* is the function

$$Q \colon I \to \mathbb{R}, \qquad Q(t) = \int_X j^0(t, x)\, \mathrm{d}^d x.$$

The unindexed $Q(t)$ denotes this integral. The indexed Q^a denotes a characteristic.

On the torus the integral always exists. Differentiation under the integral is justified by continuity on each compact slab, as in Lecture 1. For a conserved current periodicity gives

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = -\sum_{i=1}^{d} \int_{\mathbb{T}^d} \partial_i j^i \, \mathrm{d}^d x = 0.$$

Indeed, integrating in x^i gives j^i at 2π minus its value at zero. Those values agree. The charged Klein–Gordon solution above therefore has constant charge $Q = -2\pi\sqrt{13}|A|^2$.

For $\mathbb{R}^d$ an integrability hypothesis is necessary. Suppose $j^0(t, \cdot)$ is in $L^1(\mathbb{R}^d)$ for every $t \in I$. Suppose also that, for every compact interval $[t_0, t_1] \subset I$, the spatial flux satisfies

$$\int_{t_0}^{t_1} \int_{\mathbb{R}^d} \sum_{i=1}^{d} |j^i(t, x)| \, \mathrm{d}^d x \, \mathrm{d}t < \infty. \tag{2.8}$$

These conditions imply that the charge of a conserved current is constant. We prove this without differentiating an integral over all space. Choose a C^2 function $\chi \colon \mathbb{R}^d \to [0, 1]$ equal to one for $|x| \le 1$ and zero for $|x| \ge 2$. Put $\chi_R(x) = \chi(x/R)$ for $R \ge 1$. Such a function can be made from a C^2 polynomial transition between the two constant values, with its first two derivatives zero at both joins. Define $Q_R(t) = \int \chi_R j^0 \, \mathrm{d}^d x$. Compact support allows differentiation and integration by parts. The local conservation law gives

$$Q_R(t_1) - Q_R(t_0) = \int_{t_0}^{t_1} \int_{\mathbb{R}^d} (\partial_i \chi_R) j^i \, \mathrm{d}^d x \, \mathrm{d}t.$$

The absolute value of this expression is at most $\|\nabla\chi\|_\infty / R$ times the finite integral in (2.8). It tends to zero. At either endpoint,

$$|Q_R(t) - Q(t)| \le \int_{|x|>R} |j^0(t,x)|\,\mathrm{d}^d x \longrightarrow 0$$

by integrability. Taking the limit proves $Q(t_1) = Q(t_0)$. This argument states precisely how the spatial boundary contribution vanishes.

One explicit sufficient decay condition is that, on each compact time interval, there are constants $C > 0$ and $\sigma > 0$ with

$$|j^0(t,x)| + \sum_{i=1}^{d} |j^i(t,x)| \le C(1+|x|)^{-d-\sigma}.$$

The integral at large radius is bounded by a constant times $\int_1^\infty r^{-1-\sigma}\,\mathrm{d}r$, which converges.

2.5 Translations and the field equation

Suppose the density has no explicit dependence on x^ν. The point translation $x' = x + \varepsilon e_\nu$ preserves the density including its volume. Thus $b^\mu = 0$ in (2.3). Its characteristic and fixed-point divergence are $Q^a = -\partial_\nu \phi^a$ and $B^\mu = -\delta^\mu_\nu L$. Its current is

$$j^\mu = -p^\mu_a \partial_\nu \phi^a + \delta^\mu_\nu L = -T^\mu{}_\nu, \qquad T^\mu{}_\nu := p^\mu_a \partial_\nu \phi^a - \delta^\mu_\nu L.$$

Lecture 3 studies this canonical stress tensor. It chooses the negative translation $x' = x - \varepsilon e_\nu$, which gives $Q^a = \partial_\nu \phi^a$, $B^\mu = \delta^\mu_\nu L$ and current $T^\mu{}_\nu$ itself.

The sign of a translation parameter is separate from the orientation of a boundary. With $\mathrm{d}\Sigma_\mu = \iota_{\partial_\mu}\mathrm{vol}$, the flux form is $j^\mu \mathrm{d}\Sigma_\mu$. Its outward-first integrals on the top and bottom of a slab are $Q(t_1)$ and $-Q(t_0)$. Lecture 3 proves the regional balance on rectangles from the fundamental theorem of calculus.

Noether's identity holds *off shell*, meaning for arbitrary fields. The conservation law holds *on shell*, meaning on solutions. For example, keep the complex Klein–Gordon density but take $\phi_1 = 1$ and $\phi_2 = t^2$ on $\mathbb{R}\times\mathbb{T}$. Formula (2.7) gives $j^0 = 2t$ and $j^1 = 0$. Its divergence is 2. The Euler–Lagrange expressions are $E_1 = -m^2$ and $E_2 = -2 - m^2 t^2$, so

$$-E_a Q^a = -\big((-m^2)(-t^2) + (-2 - m^2 t^2)\cdot 1\big) = 2.$$

The identity remains true, but the charge $Q(t) = 4\pi t$ is not constant.

Summary. A variational symmetry satisfies $\delta_Q L = \partial_\mu B^\mu$ for arbitrary fields. Its current is $j^\mu = p^\mu_a Q^a - B^\mu$, and $\partial_\mu j^\mu = -E_a Q^a$. On solutions the divergence vanishes. The integrated charge is constant on the torus, or on $\mathbb{R}^d$ under the integrability conditions of Section 2.4. Translations give the stress tensor studied in Lecture 3.

Self-check. For the complex field, replace $e^{\mathrm{i}\varepsilon}\Phi$ by $e^{-\mathrm{i}\varepsilon}\Phi$. What happens to the characteristic and current?

Answer. Both change sign: $(Q^1, Q^2) = (\phi_2, -\phi_1)$ and $j^\mu = -\operatorname{Im}(\overline{\Phi}\partial^\mu\Phi)$. The density is still invariant.

Self-check. For the massless real density $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi$, find the current of the shift $\phi \mapsto \phi+\varepsilon$.

Answer. The characteristic is $Q^1 = 1$ and $B^\mu = 0$, so $j^\mu = \partial^\mu\phi$. Its divergence is $\Box\phi = -E[\phi]$. It is conserved on solutions of the massless equation.

Self-check. For a density with no explicit time dependence, use $\phi_\varepsilon(t,x) = \phi(t+\varepsilon,x)$ to find Q^a, B^μ and j^μ.

Answer. Here $Q^a = \partial_t\phi^a$. The chain rule gives $\delta_Q L = \partial_t L$, so $B^\mu = \delta^\mu_0 L$. Thus $j^\mu = p^\mu_a\phi^a_t - \delta^\mu_0 L = T^\mu{}_0$.

Self-check. On $\mathbb{R}\times\mathbb{R}$, consider the current $j^0 = 1$ and $j^1 = 0$. Does its local conservation law define a finite charge?

Answer. Its divergence is zero, but $\int_{\mathbb{R}} j^0\,\mathrm{d}x = \infty$. The integrability hypothesis fails, so this current has no charge as defined here.